

Interview assignment: Postdoc researcher EYE-TEACH

This assignment is grounded in two specific work packages from the EYE-TEACH project:

- **WP1:** One of the main tasks in WP1 focuses on generating guidelines and materials to support teachers in implementing and using AI-assisted eye-tracking analytics tools. This involves synthesizing findings from earlier tasks, such as teachers' technology readiness, cognitive needs, and usability, to develop best practices, design principles, and ethical guidelines. These materials will be tested with educators across multiple European countries and refined based on their feedback.
- **WP3:** WP3 tasks involve the pilot system trial implementation and evaluation, including training machine learning models to process eye-tracking metrics and produce comprehensible output for users, as well as front-end dashboard development and exploration of large language models for user interventions.

Assignment description: You are asked to complete two tasks based on the SB-SAT eye-tracking dataset, a publicly available resource containing eye movement data from participants reading SAT passages and answering comprehension questions.

- Reference Paper: [SB-SAT: A Large-Scale Eye-Tracking Dataset for Reading Comprehension](#)
- Repository: [SB-SAT GitHub Repository](#) (it includes the dataset as well as implemented methods for ML model training)

Task 1: Data exploration and analysis

Analyse the SB-SAT dataset so you get insights into meaningful patterns in the eye movement data that would be informative for teachers. You are free to suggest your own approach or can use one of the following:

Predictive modeling: Train a machine learning model to predict a meaningful outcome, for example, whether a participant answered a comprehension question correctly, or estimate the reading difficulty. *Clustering:* Perform a cluster analysis to identify distinct reading behavior profiles. *Statistical analysis:* Conduct a statistical analysis to examine relationships between eye-tracking measures and reading outcomes. This could involve regression models, mixed-effects models, or other techniques. *LLM insights:* Design (or demonstrate) a method (e.g., prompt engineering, Retrieval-Augmented Generation) to translate raw or analyzed data into pedagogical insights or instructional strategies.

Task 2: Study design & guidelines preparation

Design a small-scale evaluation study for a dashboard that would communicate the insights derived from your analysis in Part 1. Propose a study design to evaluate an aspect of the dashboard prototype with a small group of teachers.

Presentation

Prepare a **15-minute presentation** that covers both tasks. The presentation should cover:

- A summary of your approach: which features you extracted, which method you applied, and why.
- Your key findings or a concrete example of the pipeline and expected outputs (e.g., model accuracy, cluster characteristics, significant statistical relationships, or LLM prompts with example outputs).
- Your proposed study design from Task 2.

You will present this during the interview, followed by discussion. For Task 1, we do not expect a complete analysis or results. The goal is to demonstrate your ability to explore and work with an eye-tracking dataset, showing how you explore and work with such data. The emphasis is on demonstrating how you approach such a task within 15 minutes. We do not expect you to cover everything.

Submission

Please submit your presentation and code by email to konstantinos.tsiakas@ou.nl before Monday, 14 September 2026, at 23:59 CET. If you have any questions about the assignment, please email the same address.